www.PapaCambridge.com

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

*8094874271*

**MATHEMATICS (SYLLABUS D)** **4024/22**

Paper 2 **October/November 2013**

**2 hours 30 minutes**

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments
Electronic calculator

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

**Section A**
Answer **all** questions.

**Section B**
Answer any **four** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.
You are expected to use an electronic calculator to evaluate explicit numerical expressions.
If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.
For $\pi$, use either your calculator value or 3.142, unless the question requires the answer in terms of $\pi$.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 100.

This document consists of **24** printed pages.

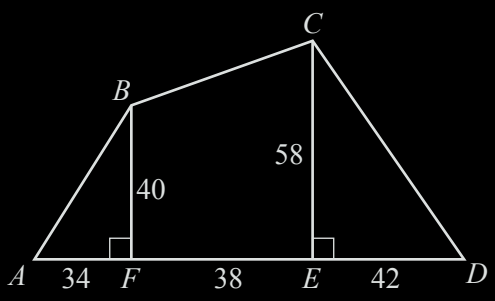

© UCLES 2013

www.PapaCambridge.com

**Section A** [52 marks]

Answer **all** questions in this section.

**1**




*ABCD* is a level field.
*F* and *E* are points on *AD* such that *BF* and *CE* are perpendicular to *AD*.
*BF* = 40 m and *CE* = 58 m.
*AF* = 34 m, *FE* = 38 m and *ED* = 42 m.

**(a)** Calculate the area of the field.

*Answer* ..... $m^2$ [3]

© UCLES 2013

www.PapaCambridge.com

**(b)** Calculate the length of $BC$.

*Answer* ..... m [2]

**(c)** Calculate $C\hat{D}E$.

*Answer* ..... [2]

© UCLES 2013

www.PapaCambridge.com

**2** **(a)** The results of a survey of the number of cars owned by 50 families are given in the table.

| Number of cars | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Number of families | 4 | 35 | 6 | 5 |

**(i)** Calculate the mean number of cars per family.

*Answer* .....[2]

**(ii)** When the same 50 families were surveyed at a later date, the results were as follows.

| Number of cars | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Number of families | $x$ | 37 | $y$ | 5 |

The mean number of cars per family stayed the same as before.

Find $x$ and $y$.

*Answer* $x =$ .....

$y =$ .....[2]

© UCLES 2013

www.PapaCambridge.com

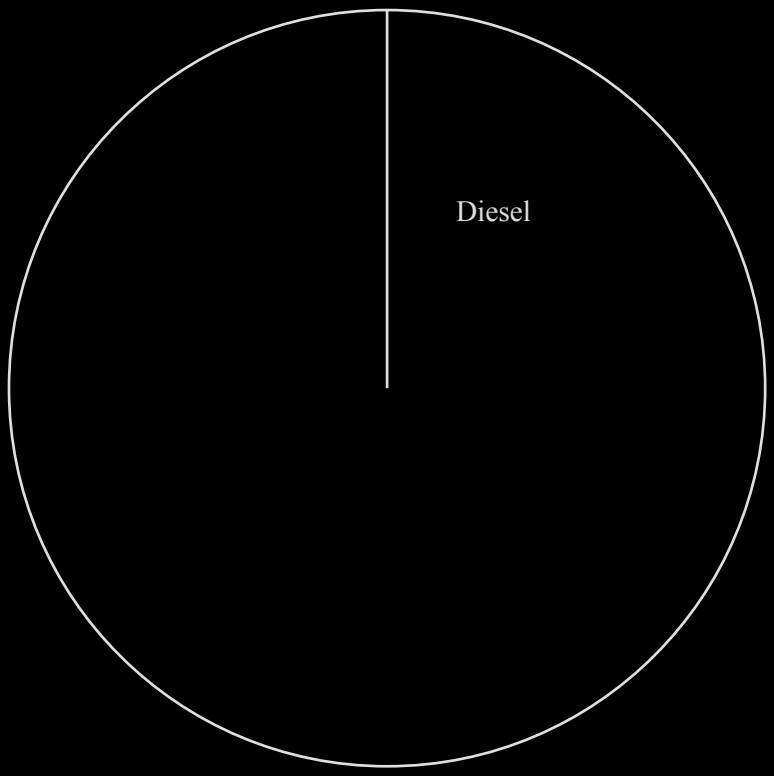
**(b)** A service station sells diesel, unleaded and super unleaded fuel.
During one week, 13 500 litres of diesel and 36 000 litres of unleaded were sold.
The total number of litres of fuel sold that week was 54 000.

**(i)** What fraction of the total number of litres sold was super unleaded?
Give your answer in its lowest terms.

*Answer* ..... [1]

**(ii)** Complete the pie chart to represent the amounts of fuel sold.

*Answer*




[3]

© UCLES 2013

www.PapaCambridge.com

**3** **(a)** Find the value of $\dfrac{a+\sqrt{a^2+b^2}}{a^2-2ab}$ when $a=-4$ and $b=-3$.

Give your answer as a fraction.

*Answer* .....[2]

**(b)** Expand the brackets and simplify $(3x^2-1)(2x+3)-x(9x-2)$.

*Answer* .....[2]

**(c)** **(i)** Factorise $9x^2+5x-4$.

*Answer* .....[1]

© UCLES 2013

www.PapaCambridge.com

**(ii)** Use your answer to **part (c)(i)** to solve the equation $9x^2 + 5x - 4 = 0$.

*Answer* $x =$ ..... or ..... [1]

**(d)** The sum of three consecutive integers is 84.

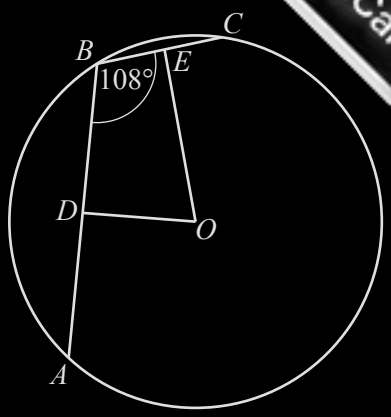
Find these three integers.

*Answer* ..... , ..... , ..... [2]

© UCLES 2013

www.PapaCambridge.com

**4** **(a)** $AB$ and $BC$ are chords of a circle centre $O$.
$D$ is the midpoint of $AB$ and $E$ is the midpoint of $BC$.
$A\hat{B}C = 108°$.

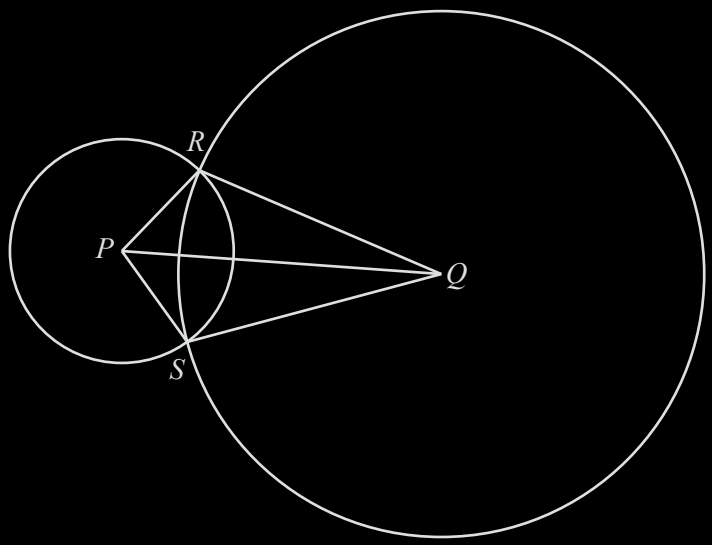



Find $D\hat{O}E$ giving your reasons.

*Answer* $D\hat{O}E$ = ..... because .....

.....[2]

**(b)**




A circle centre $P$ and a circle centre $Q$ intersect at $R$ and $S$.

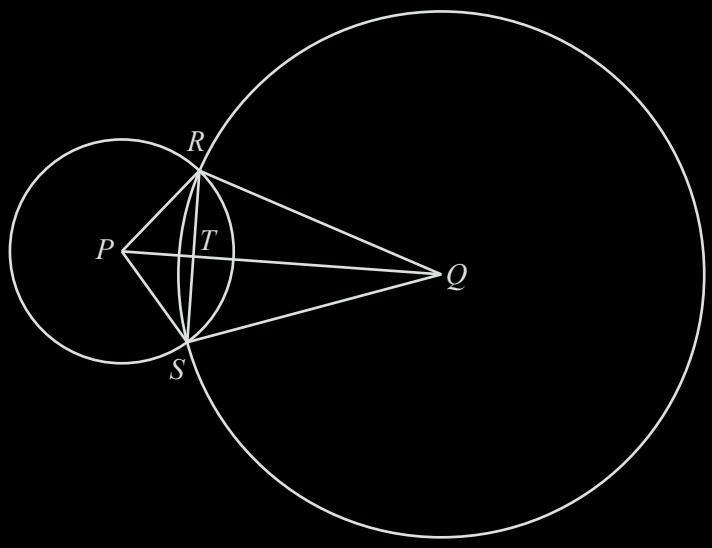
**(i)** Show that triangle $PRQ$ is congruent to triangle $PSQ$.

[3]

© UCLES 2013

www.PapaCambridge.com

**(ii)**




*RS* and *PQ* intersect at *T*.

**(a)** State the name of the special quadrilateral *PRQS*.

*Answer* ..... [1]

**(b)** Find $P\hat{T}R$.

*Answer* ..... [1]

© UCLES 2013

www.PapaCambridge.com

**5** **(a)** $\mathscr{E} = \{x : x \text{ is an integer and } 2 \leqslant x \leqslant 12\}$
$M = \{x : x \text{ is a multiple of } 3\}$
$P = \{x : x \text{ is a prime number}\}$

**(i)** $a \in M \cap P$

Find $a$.

*Answer* ..... [1]

**(ii)** Find $(M \cup P)'$.

*Answer* ..... [1]

**(b)** In a survey, 90 people were asked "Do you own a car?" and "Do you own a bicycle?".
A total of 27 people said they owned a bicycle.
Of these, 13 owned **only** a bicycle.
11 people owned neither a car nor a bicycle.

By drawing a Venn diagram, or otherwise, find how many people said that they owned a car.

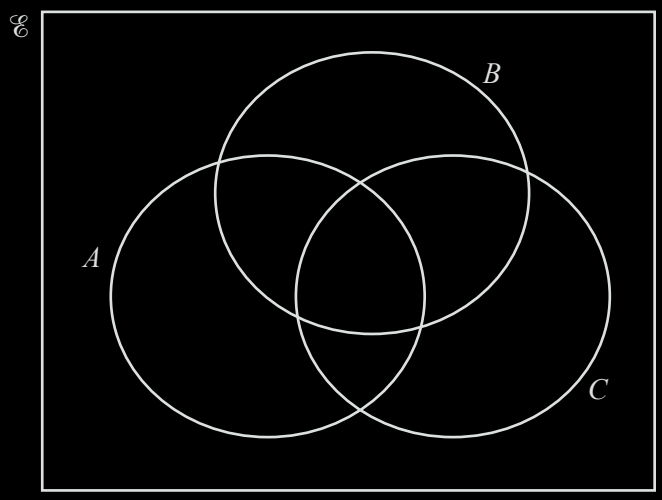
*Answer* ..... [2]

© UCLES 2013

www.PapaCambridge.com

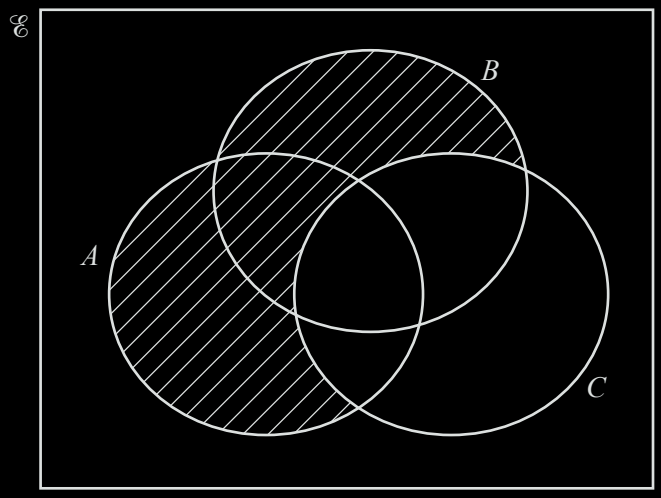
**(c)** The Venn diagrams show a Universal set, $\mathscr{E}$, and subsets $A$, $B$ and $C$.

**(i)** Shade the set $(A \cup C)' \cap B$.




[1]

**(ii)** Express in set notation the subset shaded in this diagram.




*Answer* ..... [1]

© UCLES 2013

www.PapaCambridge.com

**6** **(a)** **(i)** The cost price of bicycle A is \$620.
The shopkeeper sells it and makes a profit of 45%.

Calculate the selling price.

*Answer* \$..... [1]

**(ii)** In a sale, the price of bicycle B is reduced from \$2400 to \$1596.

Calculate the percentage reduction given.

*Answer* .....% [2]

**(iii)** Tax on the original price of bicycle C is charged at 20% of the original price.
After tax has been included, Matthew pays \$1080 for this bicycle.

Calculate the original price.

*Answer* \$..... [2]

**(b)** Ada invests \$600 in an account that earns simple interest.
At the end of 3 years, the investment is worth \$681.

Calculate the rate of simple interest per year.

*Answer* .....% [3]

© UCLES 2013

www.PapaCambridge.com

**7** **(a)** Express as a single matrix $5\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} - 4\begin{pmatrix} 1 \\ -3 \\ 0 \end{pmatrix}$.

*Answer* [2]

**(b)** Express as a single matrix $\begin{pmatrix} 7 & -1 & 3 \\ 2 & 0 & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$.

*Answer* [2]

**(c)** $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ -2 & 4 \end{pmatrix}$

**(i)** Find $\mathbf{A}^{-1}$.

*Answer* $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

**(ii)** $\mathbf{B} + 3\mathbf{I} = \mathbf{A}$ where $\mathbf{I}$ is the $2 \times 2$ identity matrix.

Find $\mathbf{B}$.

*Answer* $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

© UCLES 2013

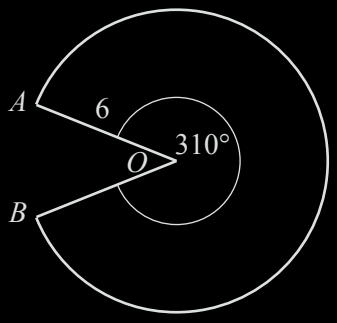
www.PapaCambridge.com

**Section B** [48 marks]

Answer **four** questions in this section.

Each question in this section carries 12 marks.

**8**




The diagram shows a sector $AOB$ of a circle with centre $O$ and radius 6 cm.
The angle of the sector is 310°.

**(a)** Calculate the total perimeter of the sector.

*Answer* ..... cm [3]

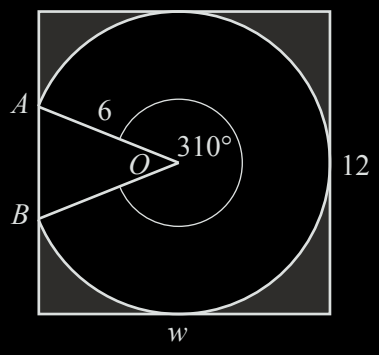
**(b)** Calculate the area of the sector.

*Answer* ..... $cm^2$ [2]

© UCLES 2013

www.PapaCambridge.com

**(c)** This sector is cut from a rectangular piece of card of height 12 cm and width $w$ cm.




One edge of the rectangular piece of card passes through $A$ and $B$.
The other edges are tangents to the circle.

**(i)** Calculate the value of $w$.

*Answer* ..... [3]

**(ii)** When the sector is cut out, the triangle $AOB$ is retained.
The rest of the rectangular piece of card, shown shaded, is discarded as waste.

Calculate the percentage of the rectangular piece of card that is discarded as waste.

*Answer* .....% [4]

© UCLES 2013

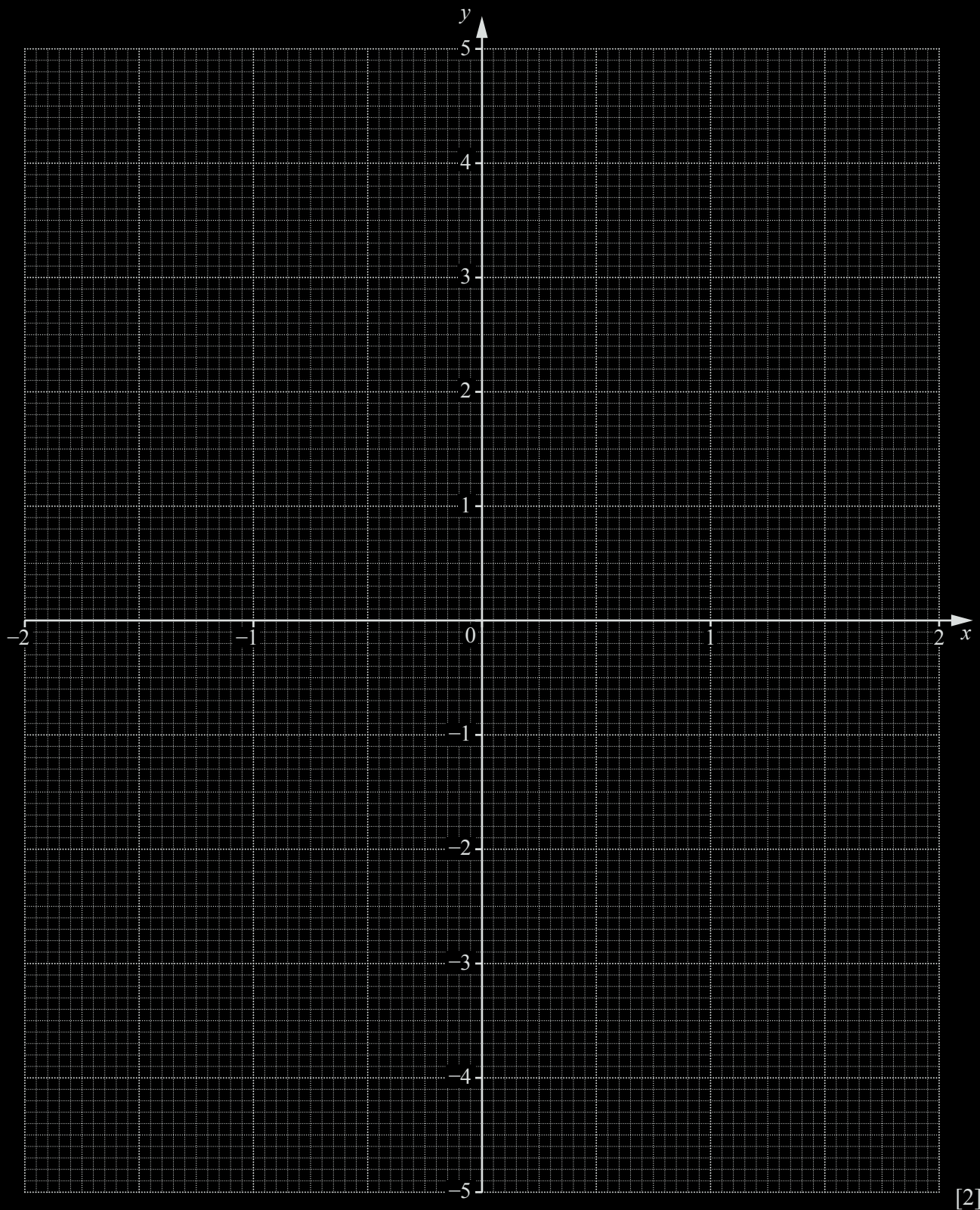
www.PapaCambridge.com

**9** The variables $x$ and $y$ are connected by the equation $y = x + \frac{1}{x}$.

The table below shows some values of $x$ and the corresponding values of $y$.
The values of $y$ are correct to 2 decimal places where appropriate.

| $x$ | 0.25 | 0.5 | 0.75 | 1 | 1.25 | 1.5 | 1.75 | 2 |
|---|---|---|---|---|---|---|---|---|
| $y$ | 4.25 | 2.5 | 2.08 | 2 | 2.05 | 2.17 | 2.32 | 2.5 |

**(a)** On the grid, plot the points given in the table and join them with a smooth curve.




[2]

© UCLES 2013

www.PapaCambridge.com

**(b)** By drawing a tangent, estimate the gradient of the curve when $x = 0.75$.

*Answer* ..... [2]

**(c)** Let $f(x) = x + \frac{1}{x}$.

**(i)** Given that $f(a) = b$, find $f(-a)$ in terms of $b$.

*Answer* ..... [1]

**(ii)** Hence, or otherwise, complete the table below for $y = x + \frac{1}{x}$.

| $x$ | –2 | –1.75 | –1.5 | –1.25 | –1 | –0.75 | –0.5 | –0.25 |
|---|---|---|---|---|---|---|---|---|
| $y$ | | | | | –2 | | | |

[1]

**(iii)** On the grid opposite, draw the graph of $y = x + \frac{1}{x}$ for $-2 \leqslant x \leqslant -0.25$. [1]

**(iv)** Write down an estimate for the gradient of the curve when $x = -0.75$.

*Answer* ..... [1]

**(d)** **(i)** On the grid opposite, draw the graph of the straight line $y = 4 - x$. [1]

**(ii)** Write down the $x$-coordinate of each of the points where the graphs of $y = 4 - x$ and $y = x + \frac{1}{x}$ intersect.

*Answer* $x$ = ..... and ..... [1]

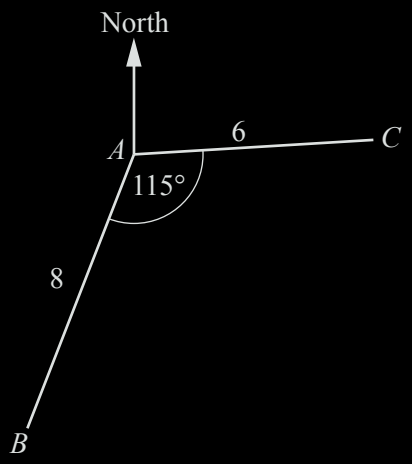
**(iii)** Find the equation for which these $x$ values are the solutions.
Give your equation in the form $Ax^2 + Bx + C = 0$.

*Answer* ..... [2]

© UCLES 2013

www.PapaCambridge.com

**10 (a)**




Two boats sail from $A$. One boat sails to $B$, and the other boat sails to $C$.
$AB = 8$ km, $AC = 6$ km and $B\hat{A}C = 115°$.

**(i)** Calculate the distance, $BC$, between the boats.

*Answer* ..... km [4]

**(ii)** The bearing of $B$ from $A$ is 200°.

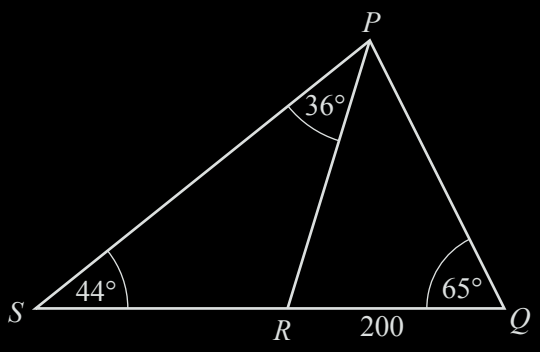
Find the bearing of $A$ from $C$.

*Answer* ..... [2]

© UCLES 2013

www.PapaCambridge.com

**(b)**




In triangle $PQS$, $S\hat{Q}P = 65°$ and $Q\hat{S}P = 44°$.
$R$ is the point on $QS$ such that $QR = 200$ m and $R\hat{P}S = 36°$.

**(i)** In triangle $PQR$, by using the sine rule, show that $PR = \dfrac{200 \sin 65}{\sin 35}$.

[2]

**(ii)** Hence show that $SR = \dfrac{200 \sin 65 \sin 36}{\sin 35 \sin 44}$.

[2]

**(iii)** Hence find the length of $SR$.

*Answer* ..... m [1]

**(iv)** Hence evaluate $\dfrac{\text{area of triangle } SPQ}{\text{area of triangle } PQR}$.

*Answer* ..... [1]

© UCLES 2013

www.PapaCambridge.com

**11** **(a)** Express as a single fraction, in its simplest form, $\frac{7}{p+2} - \frac{4}{2p-3}$.

*Answer* ..... [3]

**(b)** The distance between London and York is 320 km .
A train takes $x$ hours to travel between London and York.

**(i)** Write down an expression, in terms of $x$, for the average speed of the train.

*Answer* .....km/h [1]

**(ii)** A car takes $2\frac{1}{2}$ hours longer than a train to travel between London and York.
The average speed of the train is 80 km/h greater than the average speed of the car.

Form an equation in $x$ and show that it simplifies to $2x^2 + 5x - 20 = 0$.

[3]

© UCLES 2013

www.PapaCambridge.com

**(iii)** Solve the equation $2x^2 + 5x - 20 = 0$, giving your answers correct to 2 decimal [illegible]

*Answer* $x =$ ..... or ..... [3]

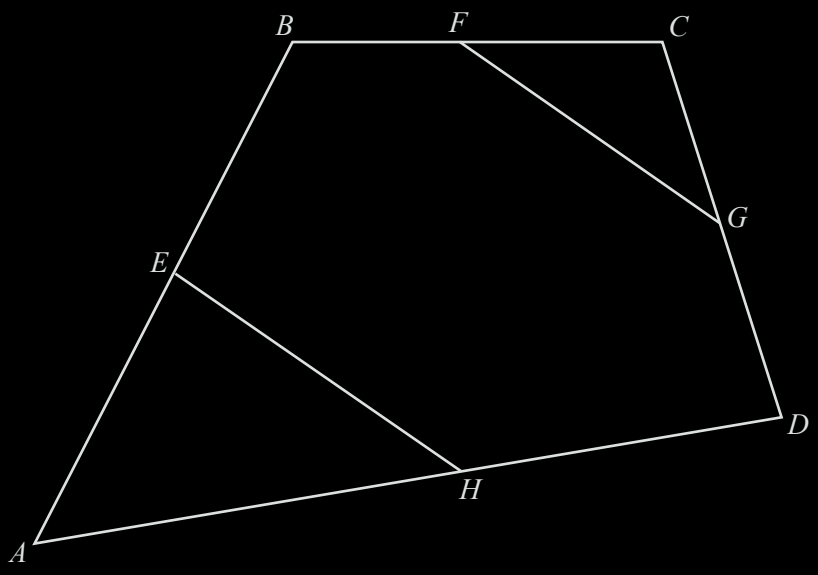
**(iv)** Hence find the average speed of the car correct to the nearest km/h.

*Answer* .....km/h [2]

© UCLES 2013

www.PapaCambridge.com

**12 (a)**




**(i)** $\overrightarrow{AD} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$

Calculate $|\overrightarrow{AD}|$.

*Answer* ..... [1]

**(ii)** $\overrightarrow{AE} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$H$ is the midpoint of $AD$.

Find $\overrightarrow{EH}$.

*Answer* $\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [2]

© UCLES 2013

www.PapaCambridge.com

**(iii)** $\overrightarrow{BF} = \begin{pmatrix} 1.5 \\ 0 \end{pmatrix}$ $\overrightarrow{CG} = \begin{pmatrix} 0.5 \\ -1.5 \end{pmatrix}$

$F$ is the midpoint of $BC$.

Find $\overrightarrow{FG}$.

*Answer* $\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [1]

**(iv)** Use your answers to **parts (ii)** and **(iii)** to complete the following statement.

The lines $EH$ and $FG$ are ..... and ..... [1]

**(v)** Given that $E$ is the midpoint of $AB$, show that $G$ is the midpoint of $CD$.

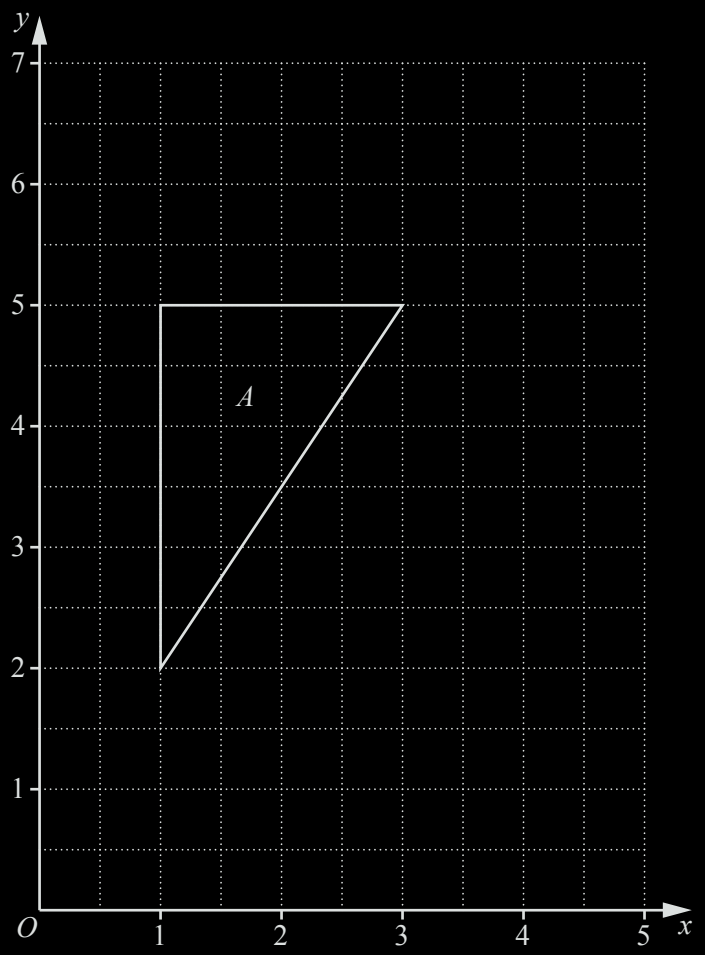
[2]

**TURN OVER FOR THE REST OF THIS QUESTION**

© UCLES 2013

www.PapaCambridge.com

**(b)**




Triangle $A$ has vertices (1, 2), (1, 5) and (3, 5).

**(i)** An enlargement, centre (1, 2), scale factor 1.5, maps triangle $A$ onto triangle $B$.

Draw triangle $B$. [2]

**(ii)** An enlargement, centre (1, 2), scale factor $-0.5$, maps triangle $A$ onto triangle $C$.

Draw triangle $C$. [2]

**(iii)** Find the ratio area of triangle $C$ : area of triangle $B$.

*Answer* ..... : ..... [1]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

© UCLES 2013